

Calculators and Communication Devices are not allowed in the examination room

Multiple Choice Questions : Circle the correct answer. No justification is required.

1. (5 Pts) The unit tangent vector at the point $(0, 2, 0)$ on the curve

$$\mathbf{r}(t) = (\tan^{-1} t) \mathbf{i} + (2e^{2t}) \mathbf{j} + (8te^t) \mathbf{k} \text{ is}$$

- (A) $\mathbf{i} + 4\mathbf{j} + 8\mathbf{k}$ (B) $\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}$ (C) $\mathbf{i} + 4\mathbf{j} + 16\mathbf{k}$
(D) $\frac{1}{9}(\mathbf{i} + 4\mathbf{j} + 8\mathbf{k})$ (E) $\frac{1}{9}(\mathbf{i} + 2\mathbf{j} + 8\mathbf{k})$

2. (5 Pts) The linearization of the function $f(x, y) = 4 + x \ln(xy - 5)$ at the point $(2, 3)$ is

- (A) $6x + 4y - 24$ (B) $6x + 4y + 24$ (C) $6x + 4y - 20$
(D) $4x + 6y + 20$ (E) $3x + 2y - 20$

3. (5 Pts) The direction of greatest rate of change of $w = 3x^2 - xy + z$ at the point $(1, -1, 6)$ is

- (A) $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ (B) $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ (C) $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$
(D) $7\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ (E) $7\mathbf{i} - \mathbf{j} + \mathbf{k}$

4. (5 Pts) If $3x^2y + z \cos y = xz^3$, then $\frac{\partial y}{\partial z}$ equals to

- (A) $\frac{3x^2 - z \sin y}{3xz^2 - \cos y}$ (B) $\frac{3x^2 + z \sin y}{3xz^2 + \cos y}$ (C) $\frac{3xz^2 - \cos y}{3x^2 - z \sin y}$
(D) $\frac{6xy - z^3}{3xz^2 - \cos y}$ (E) $\frac{6xy - z^3}{3x^2 - z \sin y}$

5. (5 Pts) The volume of the solid region bounded by the surfaces $y = x^2$, $y + z = 1$ and $z = 0$ can be found by evaluating

- (A) $\int_{-1}^1 \int_1^{x^2} (1 - y) dy dx$ (B) $\int_{-1}^1 \int_{x^2}^1 (1 - y) dx dy$ (C) $\int_0^1 \int_0^{\sqrt{y}} (1 - y) dx dy$
(D) $\int_{-1}^1 \int_{x^2}^1 (1 - y) dy dx$ (E) $\int_{-1}^1 \int_{x^2}^1 (y - 1) dy dx$

6. **(10 Pts)** Find the area of the surface with parametric equations

$$x = u^2, \quad y = uv, \quad z = \frac{1}{2}v^2, \quad 0 \leq u \leq 1, \quad 0 \leq v \leq 2$$

7. **(10 Pts)** Show that $\mathbf{F} = (e^y)\mathbf{i} + (xe^y + e^z)\mathbf{j} + (ye^z)\mathbf{k}$ is a conservative vector field, then

use this fact to evaluate $\int_{(0,2,0)}^{(4,0,3)} \mathbf{F} \cdot d\mathbf{r}$.

8. **(10 Pts)** Use the Divergence Theorem to find the outward flux of the vector field

$$\mathbf{F} = (xz^2)\mathbf{i} + (x^2y - z^3)\mathbf{j} + (2xy + y^2z)\mathbf{k}$$

across the boundary surface of the solid region enclosed by the hemisphere $z = \sqrt{4 - x^2 - y^2}$ and the xy -plane.

9. **(15 Pts)** Verify Green's Theorem for $\mathbf{F} = xy\mathbf{i} + (x^2 + y^2)\mathbf{j}$, where C consists of the line segment from $(0,0)$ to $(1,1)$ followed by the arc of the curve $x = y^2$ from $(1,1)$ to $(0,0)$.

10. **(15 Pts)** Use Stokes's Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F} = \left(x - \frac{1}{2}y^2\right)\mathbf{i} + \left(y - \frac{1}{2}z^2\right)\mathbf{j} + (z - x^2)\mathbf{k}$$

and C is the positively oriented triangle with vertices $(2,0,0)$, $(0,2,0)$ and $(0,0,4)$.

11. **(15 Pts)** Let R be the square with vertices $(0,2)$, $(1,1)$, $(2,2)$, and $(1,3)$.

Evaluate $\iint_R \left(\frac{y-x}{y+x}\right) dA$ by making an appropriate change of variables.

$$(1) \text{ (D) } \mathbf{r}'(t) = \left(\frac{1}{1+t^2} \right) \mathbf{i} + (4e^{2t}) \mathbf{j} + 8(1+t)e^t \mathbf{k} : \mathbf{T}(0) = \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{1}{9} (\mathbf{i} + 4\mathbf{j} + 8\mathbf{k}).$$

$$(2) \text{ (C) } f(x, y) = 4 + x \ln(xy - 5) \quad f_x = x \ln(xy - 5) + \frac{xy}{xy - 5} \quad f_y = \frac{x^2 y}{xy - 5}$$

$$L(x, y) = f(2, 3) + f_x(2, 3)(x - 2) + f_y(2, 3)(y - 3) = 6x + 4y - 20.$$

$$(3) \text{ (E) } \nabla w(x, y, z) = (6x - y) \mathbf{i} - x \mathbf{j} + \mathbf{k}. \quad \nabla w(1, -1, 6) = 7\mathbf{i} - \mathbf{j} + \mathbf{k}.$$

$$(4) \text{ (C) } f(x, y, z) = 3x^2 y + z \cos y - xz^3 : \frac{\partial y}{\partial z} = -\frac{f_z}{f_y} = \frac{3xz^2 - \cos y}{3x^2 - z \sin y}.$$

$$(5) \text{ (D) } V = \iint_D f(x, y) dA = \int_{-1}^1 \int_{x^2}^1 (1 - y) dx dy.$$

(6) The tangent vectors are $\mathbf{r}_u = 2u\mathbf{i} + v\mathbf{j}$, and $\mathbf{r}_v = u\mathbf{j} + v\mathbf{k}$. So, a normal vector to the surface is $\mathbf{r}_u \times \mathbf{r}_v = (2u\mathbf{i} + v\mathbf{j}) \times (u\mathbf{j} + v\mathbf{k}) = u^2\mathbf{i} + 2uv\mathbf{j} + 2u^2\mathbf{k}$. Thus

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA = \int_0^1 \int_0^2 (2u^2 + v^2) dv du = \int_0^1 (4u^2 + \frac{8}{3}) du = \boxed{4}$$

(7) We have $\text{curl } \mathbf{F} = \mathbf{O}$, the domain of $\mathbf{F}$ is $\mathbb{R}^3$, and the components of $\mathbf{F}$ have continuous partial derivatives, so $\mathbf{F}$ is conservative and thus its line integral is independent of path, that is

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A), \text{ Where } f \text{ is a potential function of } \mathbf{F}.$$

From $f_x = e^y$, $f_y = xe^y + e^z$, and $f_z = ye^z$ we obtain $f(x, y, z) = xe^y + ye^z$ as a potential

$$\text{function of } \mathbf{F}. \text{ Hence } \int_{(0,2,0)}^{(4,0,3)} \mathbf{F} \cdot d\mathbf{r} = f(4, 0, 3) - f(0, 2, 0) = 4 - 2 = \boxed{2}$$

$$(8) \iint_S \mathbf{F} \cdot \mathbf{n} dS \stackrel{\text{Div. Th.}}{=} \iiint_E \text{div } \mathbf{F} dV. \text{ We have } \text{div } \mathbf{F} = z^2 + x^2 + y^2, \text{ and}$$

$$E = \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq 2, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq 2\pi\}$$

$$\begin{aligned} \iiint_E \text{div } \mathbf{F} dV &= \iiint_E (x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^2 \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin \phi d\phi \int_0^2 \rho^4 d\rho = \boxed{\frac{64\pi}{5}} \end{aligned}$$

(9) Let C_1 be the line segment from $(0, 0)$ to $(1, 1)$, and C_2 be the arc of the curve $x = y^2$ from $(1, 1)$ to $(0, 0)$.

$$\begin{aligned} \oint_C (xy) dx + (x^2 + y^2) dy &= \int_{C_1} (xy) dx + (x^2 + y^2) dy + \int_{C_2} (xy) dx + (x^2 + y^2) dy \\ &= \int_0^1 3y^2 dy + \int_1^0 (3y^4 + y^2) dy = 1 - \frac{14}{15} = \boxed{\frac{1}{15}} \end{aligned}$$

Now, let D be the region enclosed by C

$$\begin{aligned} \iint_D \left[\frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (xy) \right] dA &= \iint_D x dA = \int_0^1 \int_{y^2}^y x dx dy \\ &= \int_0^1 \frac{1}{2} (y^2 - y^4) dy = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right) = \boxed{\frac{1}{15}} \end{aligned}$$

(10) $\oint_C \mathbf{F} \cdot d\mathbf{r} \stackrel{St., Th.}{=} \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS.$

$\text{curl } \mathbf{F} = z\mathbf{i} + 2x\mathbf{j} + y\mathbf{k}$ and we take the surface S to be the planar region enclosed by C , so S is the portion of the plane $2x + 2y + z = 4$ over $D : 0 \leq x \leq 2, 0 \leq y \leq 2 - x$.

Since C is oriented counterclockwise, we orient S upward.

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \iint_D \left(-P \frac{\partial z}{\partial x} - Q \frac{\partial z}{\partial y} + R \right) dA, \text{ where } z = 4 - 2x - 2y \\ &= \iint_D (2z + 4x + y) dA = \iint_D (8 - 3y) dA \\ &= \int_0^2 \int_0^{2-x} (8 - 3y) dy dx = \int_0^2 (10 - 2x - \frac{3}{2}x^2) dx = \boxed{12} \end{aligned}$$

(11) The terms $y - x$ and $y + x$ in the integrand suggests the new variables $u = y - x$ and $v = y + x$. Solving for x and y , we get $T : x = \frac{1}{2}(-u + v), y = \frac{1}{2}(u + v)$.

The Jacobian of T is $J = \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{1}{2}.$

To find S corresponding to R we note that the sides of R lie on the lines: $y - x = 0$, $y - x = 2$, $y + x = 2$, and $y + x = 4$, so the image lines in uv -plane are: $u = 0$, $u = 2$, $v = 2$, and $v = 4$.

Thus S is given by $S : 0 \leq u \leq 2, 2 \leq v \leq 4$. The integration now follows

$$\iint_R \left(\frac{y - x}{y + x} \right) dA = \int_0^2 \int_2^4 \left(\frac{u}{v} \right) \left| -\frac{1}{2} \right| du dv = \frac{1}{2} \int_0^2 u du \int_2^4 \frac{dv}{v} = \boxed{\ln 2}$$